ELSEVIER

The Science of the Total Environment 169 (1995) 123–128

the Science of the Total Environment
An International Journal for Scientific Research into the Environment and Its Relationship with Man

Driving patterns and emissions from different types of roads

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Received 27 February 1994; accepted 30 October 1994

Abstract

The project evaluates the relationship between emissions and travel speeds on different types of roads such as city streets, highways, express roads and motorways. Approximately 800 measured driving patterns of 13 streets and roads have been analysed and emissions have been predicted in an emission model, also taking into account deterioration factors and cold start emissions. The result of the analysis is a clear relationship between travel speed (trip length divided by trip time) in the range of 10–120 km/h and emissions from all vehicle types. For petrol-powered passenger cars catalysts reduce HC, CO and NO_x emissions by 70–80% on main roads and by 60–75% on city streets. The difference is due to the proportion of cold engines in city traffic. In city streets, when cars with cold engines are included, the emissions of CO and HC from petrol-powered passenger cars are found to be 10–20% and 5–10% higher, respectively. Travel speed — and not the type of road — is crucial to the level of emissions. However, express roads have slightly higher emission levels than motorways at similar travel speeds, presumably because traffic flows are less steady on express roads than on motorways.

Keywords: Driving patterns; Cold start emissions; Travel speed; Road type; Petrol-powered passenger cars; Catalytic convertors

1. Introduction

The Danish Environmental Protection Agency and the Danish Road Directorate conducted in 1989/1990 a project that evaluated relationships between driving patterns, emissions and traffic management measures in four streets in Copenhagen with heavy traffic. The project showed a strong relationship between travel speed and emissions. The travel speeds in the four streets were between 10–60 km/h [1][1].

This project [2][2] is a follow-up study of the above mentioned. It includes streets in a middle-sized town outside Copenhagen as well as roads

[1] In Danish with an English summary. Air Pollution from Individual and Public Transportation, Danish Environmental Protection Agency, Copenhagen.

[2] In Danish with an English summary, Driving Patterns and Air Pollution — in the Provinces, The Danish Road Directorate, Copenhagen.

SSDI 0048-9697(95)04640-Z

on the main road network. The objective of the study is to examine whether or not the strong relationship between travel speed and emissions is valid for travel speeds > 60 km/h and for other types of roads. Another aim of the study is to analyze the relationship between emissions and steady and unsteady traffic flows on main roads. Travel speed is defined as trip length divided by trip time, including stops.

The Danish Road Directorate has financed the project, Spencer C. Sorenson, Laboratory for Energetics, Technical University of Denmark, 2800 Lyngby, Denmark has developed the emission model and COWIconsult has carried out the analysis.

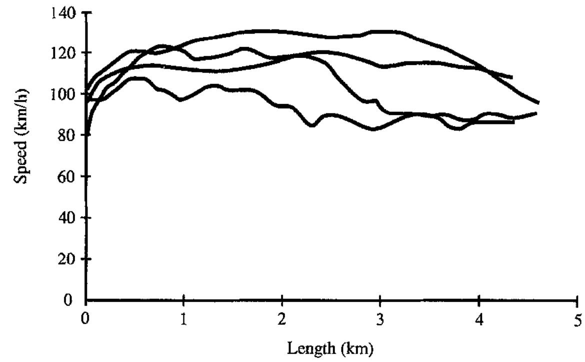

Fig. 1. Examples of driving patterns of four passenger cars on a motorway.

2. Methodology

2.1. Driving pattern

A total of ~ 800 measured driving patterns of 13 streets and roads have been registered and analyzed. During 1991 driving patterns were registered by a measuring car which followed selected vehicles. The velocity of the selected vehicle was registered every second. Based on this information a driving pattern can be established which describes the speed profile of the vehicle on the section of the road in question (Fig. 1). Six streets in the city of Roskilde outside Copenhagen represent streets with different traffic flows in a middle-sized town (50 000 inhabitants). On the main road network, seven roads have been analyzed, representing steady and unsteady traffic flows on two motorways, two express roads and three highways. At the time of the study the speed limit on city streets was 50 km/h and on highways, express roads and motorways 80 km/h, 90 km/h and 100 km/h, respectively. During 1992 the speed limit on motorways was raised to 110 km/h. The average length of the sections analyzed was 1.5 km for city streets and 6.8 km for main roads.

Furthermore, a survey of the proportion of traffic which have cold engines was carried out for the six city streets assuming no cold engines on main roads.

An existing emission model was extended to cover energy consumption and travel at high speeds. The emission model is developed by the Laboratory for Energetics, Technical University of Denmark [3]. The model predicts emissions and fuel consumption in g/km for HC, CO, NO_x and particles for a given driving pattern for the following vehicles: petrol-powered passenger cars with or without catalyst, diesel-powered passenger cars and diesel-powered heavy duty lorries (10 and 26 tonnes). Emissions and fuel consumption of light duty lorries are estimated from emissions and fuel consumption of passenger cars assuming that light duty lorries have 50% higher emissions than passenger cars due to differences in weight and size.

Emissions are predicted in the following way. The driving pattern is divided into driving conditions which can be characterized as either acceleration, deceleration, constant speed or idle. The driving condition that fits best in an established database of emissions related to different driving conditions is used and emissions can be calculated for the entire driving pattern. For petrol-powered passenger cars the emission model is based on a database established by the Environmental Protection Agency in Studsvik, Sweden (SNV) which contains many instantaneous measurements made during the different portions of various standard emission test cycles which cover speeds up to 100 km/h. The database for Swedish passenger cars has been adjusted to Danish conditions. Details on how the emission database is established for vehicles other than passenger cars are given in [3].

The emission model predicts emissions and fuel consumption from a new and warm engine. Since emissions change as a function of mileage driven for petrol-powered passenger cars, predicted emissions of HC, CO and NO_x are adjusted according to deterioration factors. Furthermore, as a cold engine has higher emissions than a warm engine for petrol-powered passenger cars, predicted emissions of HC and CO as well as energy consumption are also adjusted according to cold start emissions.

The emission model has been tested using the standard emission test cycles as driving patterns and the variations were within 5–15% which is found to be satisfactory.

2.2. Modelling emissions at high speeds

Detailed emission measurements with acceleration and deceleration are not available for passenger cars at speeds > 100 km/h. Data has been obtained for a few vehicles for constant speed operations up to 120 km/h from the Swedish database. Only integrated emission factors are available for speeds with accelerations up to 140 km/h averaged over a large number of vehicles from Technischer Überwachnungs-Verein Rheinland, Germany (TÜV Rheiland) [4]. This German study shows a significant increase in measured emissions at high speeds as emissions of HC, CO and NO_x increase by factors of 2.3, 5.2 and 2.7, respectively, from a highway cycle with a mean speed of 78 km/h to a motorway cycle of ~ 116 km/h. The measured emissions are for new catalyst cars with warm engines. The levels for HC, CO and NO_x are 0.06 g/km, 0.98 g/km and 0.28 g/km, respectively, for the highway cycle and 0.14 g/km, 5.13 g/km and 0.75 g/km, respectively, for the motorway cycle.

Therefore, to predict emissions at speeds > 100 km/h it has been necessary to extrapolate from present conditions, although this is a very uncertain procedure. Extrapolation has been carried out in a fashion which has at least some basis in technical principles as it is assumed that vehicle power is a correlating variable for higher speeds. Extrapolation has been carried out based on the Swedish database for lower speeds, converting each speed-acceleration condition to the equivalent vehicle power, for details see [3]. The German measured emissions at high speeds were found to be 3–5 times higher when compared with the Swedish extrapolated emissions at high speeds for petrol-powered passenger cars equipped with catalysts. Therefore, it is clear that other factors must be influencing the results.

An explanation could be found in the electronic control system for vehicles with three-way catalyst control. The driving cycles currently in use for emission certification all have maximum speeds < 100 km/h, and weak accelerations with the higher speeds. The manufacturers do not need to pay attention to the emissions at these conditions in order to meet emission certification test procedures. Vehicles using three-way catalysts may change to a rich engine operating mode at the highest speeds to perform accelerations at high speeds. This is especially true for smaller cars with little extra power to give away. In addition to preventing the catalyst from overheating, emissions increase significantly [3].

Therefore, two alternative calculations have been carried out for passenger cars with catalysts at high speeds. 'Cat Low', where the catalyst is assumed to be functioning at all speeds, has extrapolated data for speeds > 100 km/h based on the Swedish database for constant speed operations up to 120 km/h. 'Cat High', where the catalyst is assumed to operate in an unregulated mode at high speeds, is based on the assumption that vehicle power is a correlating variable for higher speeds, extrapolating the Swedish database for lower speeds to high speed conditions and at the same time adjusting the level of modelled emissions to the German measured integrated emissions at high speed. Since the German measured emissions are for new catalyst cars with warm engines, these emissions have further been adjusted to an average Danish car according to deterioration factors.

A small follow-up study was carried out to test the 'Cat Low' and 'Cat High' issue. The exhaust emissions and fuel consumption were measured for three petrol-powered passenger cars with catalysts registered in Denmark according to the exhaust emission standards of October 1990 when catalysts became a requirement. All cars had an engine size of 1.4 l. The tests were carried out at a test centre in Sweden. In addition to standard regulatory driving cycles the cars were also tested at constant speeds from 85 to 135 km/h and over a special motorway driving cycle. No clear conclusions could be made from the study since the average CO and HC emissions were closer to the 'Cat High' alternative while NO_x emissions were closer to the 'Cat Low' alternative [5].

3. Results

3.1. Emissions and travel speed

The result of the analysis is that there is a clear relationship between emissions and all travel speeds. The strong relationship between emissions and travel speeds for the four streets in Copenhagen is also valid for higher speeds and for city streets in a middle-sized town and for main roads (Fig. 2). However, at high speeds the predicted emissions are uncertain due to few available measured emissions at high speeds.

HC and CO emissions decrease when travel speeds increase for all types of vehicles for travel speeds up to 80–90 km/h. Above 90 km/h emissions increase. For passenger cars the relationship between NO_x emission and travel speeds is parabolic with a minimum at ~60 km/h. For heavy duty lorries a minimum is reached at travel speeds between 60–70 km/h. Energy consumption follows the same trends as NO_x emission. However, at high speeds the increase is weaker and at low speeds it is stronger. Emissions of particles have only been predicted for diesel-powered vehicles because data concerning emissions from petrol-powered vehicles is insufficient to determine emissions of particles as a function of travel speeds. For diesel-powered passenger cars the relationship between emissions of particles and travel speeds is parabolic with a minimum at 70–80 km/h. For heavy duty vehicles the relationship is less clear. However, emissions seem to decrease when travel speeds increase. In general, emissions and fuel consumption are at a minimum at travel speeds of 60–80 km/h.

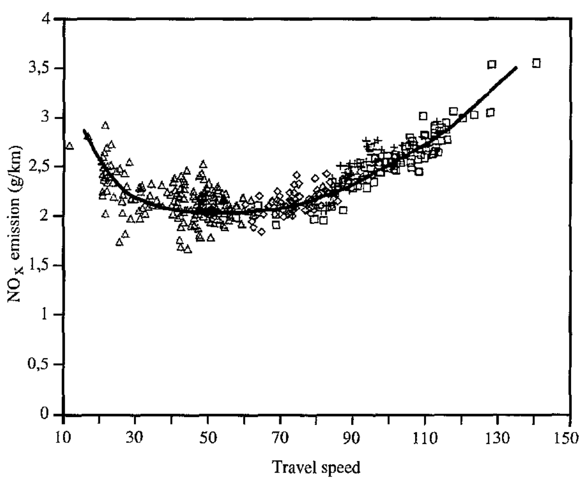

Fig. 2. NO_x emissions from passenger cars without catalysts as a function of travel speed. One dot represents a driving pattern.

Emissions at low travel speeds were found to be relatively high due to the variations in speed at low travel speeds in city streets.

For petrol-powered passenger cars catalysts reduce HC, CO and NO_x emissions by 70–80% on main roads and by 60–75% in city streets, provided the catalyst is functioning at high speeds ('Cat Low'). In the case of 'Cat High' for high speeds, reductions are considerably less on main roads. On main roads catalysts are more effective at reducing emissions of HC, CO and NO_x than in city streets, due to the fact that there are no or few cold engines on main roads (Figs. 3 and 4).

A cold engine has higher emissions of HC and CO as well as higher energy consumption than a warm engine for petrol-powered passenger cars. In city streets, when cold engines are included, the emissions of CO and HC from petrol-powered passenger cars are found to be 10–20% and 5–10% higher, respectively, than when only warm engines are considered, on the condition that 4–8% of all petrol-powered passenger cars have cold engines. Cold engines of petrol-powered passenger cars equipped with catalysts have 20–40% higher emissions of CO and 20–50% higher emissions of HC than warm engines. Cold engines of petrol-powered passenger cars without catalysts have 3–4 times higher emissions of CO and HC than warm engines. For petrol-powered passenger cars with catalysts emissions are 10–20

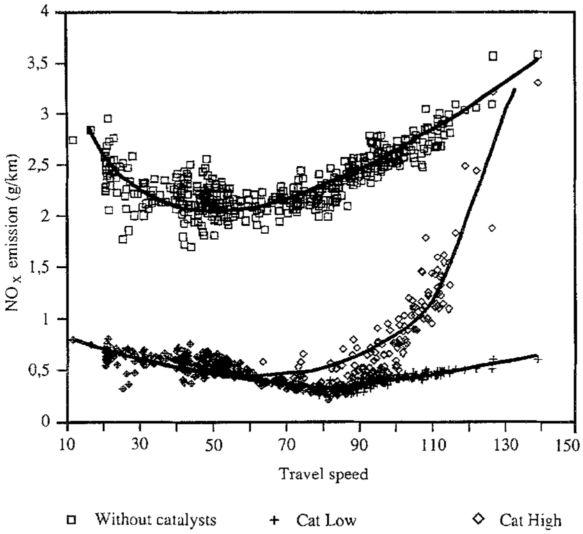

Fig. 3. NO_x emissions from passenger cars with and without catalysts as a function of travel speed.

times more than those of warm engines. A follow-up study has further analyzed the impact on national emissions of cold engines. This study concludes that although driving with cold engines accounts for only 9% of the time, it will generate 60% of HC and CO emissions when all passenger cars have catalysts [6].

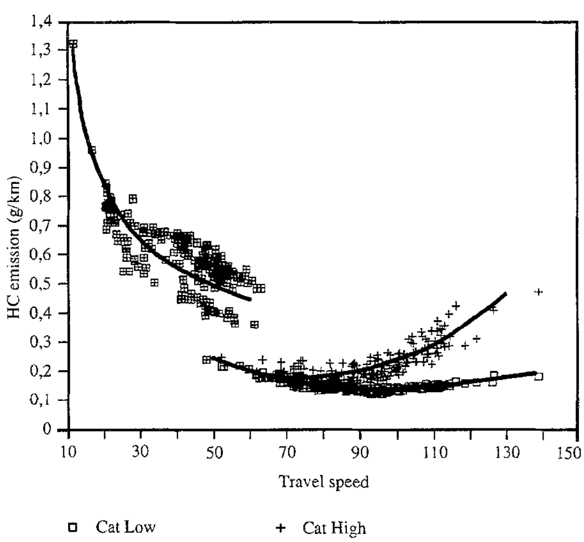

Fig. 4. Illustration of the impact of cold start emission for passenger cars with catalysts. HC emissions are substantially higher for city streets including cold engines than for main roads assuming no cold engines at the same travel speeds (50–60 km/h).

During the project period there was a debate about speed limits in Denmark. A tentative assessment of the effect of higher speed limits was carried out. Based on the relationship between travel speeds and emissions the effects of increased travel speeds from 80 km/h to 100 km/h and from 100 km/h to 110 km/h have been estimated. HC emissions will increase marginally and CO emissions slightly.

Increases of NO_x emissions will by and large follow the increases of petrol-powered passenger cars, that is, ~ 15% from 80 km/h to 100 km/h and ~ 10% from 100 km/h to 110 km/h ('Cat Low'). Increases will be higher in the 'Cat High' alternative (Figs. 3 and 4). Increased travel speeds will only increase emissions of particles slightly. Energy consumption will increase along the lines of NO_x emissions from passenger cars, that is, ~ 14% from 80 km/h to 100 km/h and ~ 8% from 100 km/h to 110 km/h. Eventually, the speed limit on motorways was raised from 100 km/h to 110 km/h and remained 80 km/h on highways and 90 km/h on express roads.

3.2. Emissions and different types of roads

The relationship between emissions, types of roads and traffic flows has been illustrated by the example of NO_x emissions from petrol-powered passenger cars without catalysts. In general, the relationship between emissions and type of road shows that travel speeds and not the type of streets or roads are crucial to the level of emissions. However, NO_x emissions on express roads showed 6% higher emissions than on motorways for a travel speed of 80 km/h dropping to 0% for a travel speed of 140 km/h, presumably because traffic flows are less steady on express roads than on motorways (Fig. 5). Based on qualitative assumptions main roads with steady and unsteady traffic flows on motorways, express roads and highways were selected. The analysis of the relationship between emissions and travel speeds showed no significant difference between steady and unsteady traffic flows for the same travel speeds, indicating that driving patterns are much alike for the same travel speed regardless of the type of road. A follow-up study has further analyzed the relationship between speed fluctuations and travel speeds. This study concludes that travel

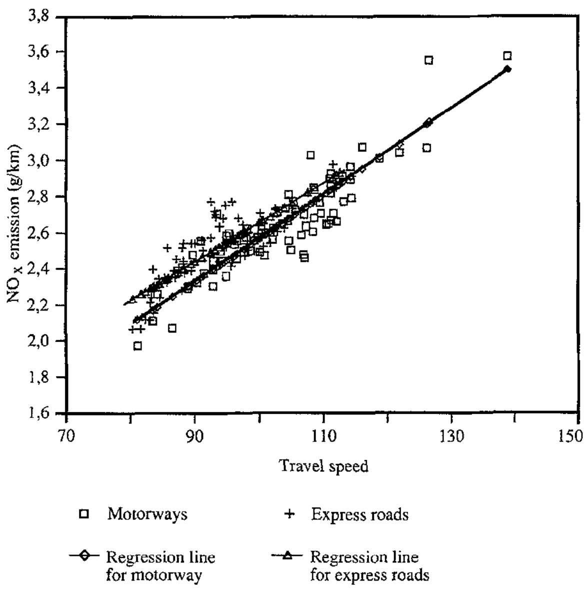

Fig. 5. NO_x emissions from passenger cars without catalysts as a function of travel speed on motorways and express roads. Regression lines for motorways and express roads are shown.

speed is the most important factor with speed fluctuation being of a lesser importance in relation to emissions [7].

3.3. Total emission loads

For the selected motorways, express roads, highways and city streets the total emission loads have been calculated based on average emission factors of the different types of vehicles and their numbers. HC and CO emissions per km travelled are higher for city streets than for main roads, while NO_x emissions and energy consumption per km travelled are higher on main roads compared with city streets. The distribution of emission load by type of vehicle has been calculated. Approximately 85–90% of HC and CO emissions come from passenger cars, whereas 50% of NO_x emissions originate from heavy duty traffic. On main roads, 50% of emissions of particles are derived from heavy duty traffic compared with 60% in city streets. In city streets light duty traffic causes ~ 25% of the total load of emissions of particles. Estimates are based on the 'Cat Low' alternative and on the assumption that 9% of the traffic flow in 1991 was petrol-powered passenger cars equipped with catalysts.

4. Discussion

The study revealed the need for further measurements of emissions at high speeds in order to predict reliable emissions based on driving patterns at high speeds. Furthermore, the impact of cold engines on emissions is significant for city streets and especially with a growing proportion of cars equipped with catalysts. Measures to reduce the impact of cold engines are essential.

The relationship between travel speeds, and emissions and energy consumption for all types of vehicles has gained widespread application in Denmark. Municipalities carry out mapping of the environmental impact of traffic in the preparation of local action plans based on these relations using spread sheet models or traffic models with environmental impact mapping facilities. Furthermore, the relationship between travel speeds, and emissions and energy consumption is used to carry out environmental impact assessments (EIA) of, for example, motorways, etc.

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